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ORIGINAL PAPER

Quantitative Parametrization of Mixture Distribution in GDI Engines: A CFD Analysis

S. Krishna Addepalli[1] · J. M. Mallikarjuna[1]

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Abstract

This paper presents an objective classification of mixture distribution in the combustion chamber of a gasoline direct injection (GDI) engine into homogeneous and non-homogeneous types. The non-homogeneous mixture distribution is further classified as properly stratified, improperly stratified and mal-distributed types. Based on this classification, four types of properly stratified mixture distributions viz., random, linear, Gaussian and parabolic are virtually simulated in the combustion chamber of a GDI engine using computational fluid dynamics to identify the mixture that results in maximum indicated mean effective pressure (IMEP). It is found that the IMEP is highest for the parabolic mixture distribution which is followed by Gaussian, linear and random types. The performance and emission characteristics of the virtual mixture distributions are compared with a late fuel injection case at different over all equivalence ratios ranging from 0.3 to 0.7. Then the variation of mixture equivalence ratio with the distance from the spark plug is parametrized for different virtual mixture distribution cases and expressed using a parameter called the "stratification index". It is found that the stratification index based on Gaussian variation gives maximum information about the mixture distribution in the combustion chamber. Finally the stratification index of different virtual mixture distributions is compared with the late fuel injection case at various overall equivalence ratios. It is found that the late fuel injection case tends to produce highest IMEP when the stratification index is close to unity.

Abbreviations

BDC	Bottom dead center
CAD	Crank angle degree
CFD	Computational fluid dynamics
CFL	Courant Fredrich and Lewis
DISI	Direct injection spark ignition
ER	Equivalence ratio
EVC	Exhaust valve closing
EVO	Exhaust valve opening
GDI	Gasoline direct injection
HC	Hydro carbons
IC	Internal combustion
IMEP	Indicated mean effective pressure
IVC	Intake valve closing
IVO	Intake valve opening
Kh-rt	Kelvin Helmholtz–Rayleigh–Taylor
NO_x	Nitric oxides
PFI	Port fuel injection
PISO	Pressure implicit with the splitting of operators
RNG	Renormalized group
SI	Stratification index
SOR	Successive over relaxation
TDC	Top dead center
TKE	Turbulent kinetic energy

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1 Introduction

With a rapid growth in the usage of optical diagnostic techniques for IC engine research, many complicated phenomena like spray dynamics, mixture formation, flame development and propagation are widely studied [1–5]. However, the existing experimental techniques are not entirely sufficient to understand all the intricacies of very high-speed processes that occur in IC engines. Thus the engine researchers also depend on CFD to gain a deeper understanding of the physical processes in the IC engines [6–9]. The main advantage of CFD especially when applied

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to high-speed processes is that the simulated physical phenomenon can be understood to a greater level of detail, unlike the experimental results [10]. Also, the simulation techniques allow the researchers to study the effect of many design variations on the expected outputs without incurring extra cost, unlike the experiments [11, 12].

Many researchers developed detailed submodels to simulate physical processes that occur in IC engines like in-cylinder flows, turbulence, spray and mixture formation, combustion etc. [13–17]. The CFD sub-models are developed to imitate the natural physical processes to the extent possible and thus there are bound to be certain approximations. As the CFD technique depends on numerically and iteratively solving the governing equations of the physical phenomenon or the models, there is always a scope for numerical truncation errors to creep in [18]. Thus the results from the CFD analysis have to be validated thoroughly with the experimental results before using them [19]. Also, the results of CFD analysis are only as good as the inputs given by the user. These inputs are generally given as boundary and initial conditions which are collected from experiments. Thus CFD and the experimental techniques complement each other.

The primary goal of today's engine research is to meet the stringent emission norms that are laid by various government agencies from time to time at different parts of the world, without compromising on the power output [20]. In order to achieve this goal, CFD techniques are extensively used in the design stage of various engine development programs. Even though, the use of after-treatment devices is an effective way to reduce exhaust pollutants, it would be better to reduce pollutants formation at the source itself [21]. Operating IC engines at lean air–fuel mixtures not only conserves fuel but also reduces exhaust pollutants. Today, GDI engines are gaining popularity because of their capability to achieve stable combustion with lean air–fuel mixtures over a wide range of engine operating conditions [22–24]. The lean engine operation, in the case of GDI engines, demands mixture stratification in the combustion chamber i.e., a combustible mixture at the spark plug location at the time of spark, while a very lean mixture at the other locations [25, 26]. The mixture formation in a GDI engine is controlled by the fuel injection timing. Early fuel injection during suction stroke results in the homogeneous mixture, whereas late fuel injection during the compression stroke generates a stratified mixture. The early injection is used during high-load conditions, whereas the late injection is used during idling and mid-load conditions [25–28]. Thus, a GDI engine demands different mixture preparation strategies during its entire operating range, which is complicated, compared to its PFI counterpart [19].

Some of the most common ways to express the mixture distribution in the engine cylinder was through equivalence ratio contour plots [29–31], or by the comparison of heat release rates [32, 33], in-cylinder pressures, exhaust emissions [34, 35] and uniformity indices. Millo et al. [36] developed a mixture evaluation technique for GDI engines using the CFD analysis which allowed a preliminary assessment of the mixture quality for different fuel injection systems viz., swirl and multi-hole injectors. Their methodology predicted soot formation and oil dilution risks based on the liquid fuel mass impacting on the cylinder wall and engine valves. The developed methodology was used as a guideline to select the fuel injection systems for GDI engines.

In spite of the fact that the mixture distribution in GDI engines is comprehended to impact combustion, performance and emission characteristics significantly, there isn't much work reported in the literature on which sort of mixture distribution qualifies to be called as a good mixture. It is to be noticed that such portrayal of good mixture distribution isn't conceivable utilizing current experimental strategies in view of the trouble engaged with measuring the mixture equivalence ratio instantaneously in various areas of the combustion chamber. Therefore in this investigation, four types of mixture distributions viz. (1) random, (2) linear, (3) Gaussian and (4) parabolic, are contrasted to identify the mixture that results in highest IMEP. However the above mentioned virtual mixture distributions cannot be achieved in real engine scenarios. Hence the results of the above four mixture distributions are compared with a late injection strategy to understand the latter's deviation from the virtual cases. Finally the deviation between the real and virtual cases is quantified by parametrizing the variation of mixture equivalence ratio with the distance from the spark plug. Such quantification can be used to generate objective functions by the engine researchers to perform the optimization of mixture distribution in the combustion chamber.

2 The CFD Analysis

2.1 The Geometrical Model of the Engine

In this study, a four-stroke wall guided GDI engine with the bow-ditch piston is considered for the CFD analysis. The detailed engine specifications are shown in Table 1. The geometric model of the engine has been developed using CREO, which is shown in Fig. 1a. The engine specifications are similar to the one used by Costa et al. [37]. The bow-ditch piston is replaced by a pentroof piston with an offset bowl, as shown in Fig. 1b, in order to improve the mixture preparation at part loads. The spark plug is centrally located, whereas the fuel injector is mounted in between the intake ports.

Table 1 Engine specifications (Costa et al. [37])

Stroke	79 mm
Bore	81.3 mm
Connecting rod	143 mm
Compression ratio	10.6:1
Number of valves	4
Exhaust valve open (EVO)	27° before BDC
Exhaust valve close (EVC)	0° after TDC
Inlet valve open (IVO)	3° before TDC
Inlet valve close (IVC)	36° after BDC
Fuel system	Direct injection, 100 bar

2.2 The CFD Methodology

In this study, the CFD analysis is carried out using the CONVERGE CFD software. A cut-cell Cartesian mesh is generated during run time for the given geometry. The base grid size used is kept at 4 mm. The permanent grid embedding in the regions of valve seats, cylinder and fuel injection path, changes the grid size in these locations between 0.125 and 1 mm. Apart from this, the adaptive grid refinement is used to capture the velocity, temperature, and species concentration gradients, which reduces the mesh size to a minimum of 0.25 mm in appropriate regions. The cell count varies between 0.4 (at TDC) and 1.1 million (at BDC). A typical grid at a particular time-step at the time of ignition is shown in Fig. 2 which also shows the mesh refinement.

Initially, the CFD simulation is carried out for four consecutive cycles starting from the EVO (27 CAD before BDC) without fuel injection and combustion. All the further simulations presented are initialized using the pressure and temperature data in the respective regions that are obtained at the end of the fourth cycle. The atmospheric pressure of 1 bar and temperature of 310 K are used as the boundary conditions at the inlet and outlet because the engine considered is a naturally aspirated one [38]. The time-step is varied in between 10^{-8} and 10^{-5} s. The variation of the time-step is based on convection and diffusion

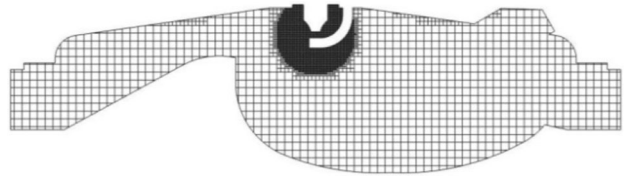


Fig. 2 A typical mesh of the combustion chamber during ignition

CFL numbers which are limited to 1 and 2 respectively. The pressure–velocity coupling is carried out using the PISO algorithm that is applied to a collocated grid using the Rhie-Chow scheme [39]. The flow turbulence is analyzed using the RNG k-ε model [40] model. It is initialized with a turbulence intensity of 5% of the inlet velocity and the length scale is assumed to be 1 mm [38, 41]. The gasoline injection process is simulated using the standard droplet discrete model and the spray breakup is modelled using the kh-rt model [42–44]. The initial droplet size is calculated using the blob injection model i.e., the initial droplet size is assumed to be equal to the nozzle exit diameter. The fuel turbulent dispersion is analyzed using the O'Rourke model [45]. The drop evaporation is analyzed using the Frossling evaporation model [45]. The NTC collision model is used to analyze droplet collisions [46] and the droplet wall interaction is analyzed using the O'Rourke wall collision model [47].

The detailed chemical kinetics during the ignition and combustion are solved by using the SAGE combustion model. In this study, iso-octane (C_8H_{18}) is used as a gasoline surrogate. The chemical kinetic mechanism developed by Givler et al. [48] is used for combustion analysis. The detailed chemistry solver, the SAGE, calculates the reaction rates for each elementary reaction while the CFD solver solves the transport equations. The mechanism contains 152 reactions and 48 species. The ignition event is modelled by introducing a high-temperature source between the spark plug electrodes at the time of ignition. The next section describes various governing and model equations that are solved during the CFD analysis.

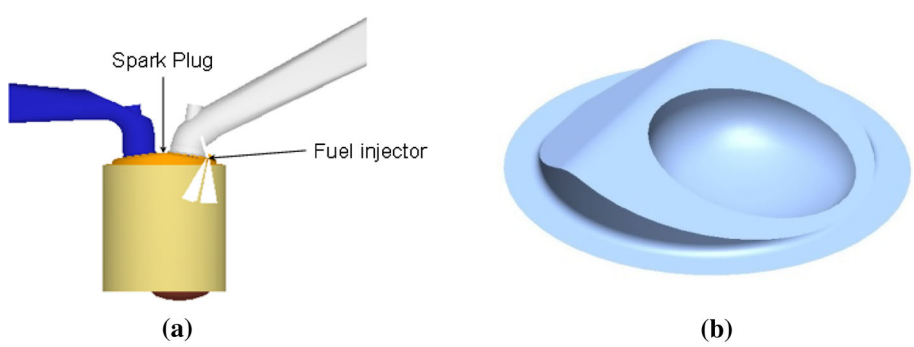


Fig. 1 The geometric model of the engine. **a** Computational domain. **b** Piston top surface

2.3 The Governing Equations

2.3.1 Mass Flow Rate and Momentum

The mass and momentum equations used to model the in-cylinder flows are given by Eqs. (1) and (2) [18].

$$\frac{\partial \rho}{\partial t} + \frac{\partial(\rho u_j)}{\partial x_j} = s_m \tag{1}$$

$$\frac{\partial \rho u_i}{\partial x} + \frac{\partial(\rho u_j u_i - \tau_{ij})}{\partial x_j} = \frac{\partial p}{\partial x_i} + S_i \tag{2}$$

where t is the time, x_i is Cartesian coordinate ($i = 1, 2, 3$), u_i is the absolute velocity component in the direction, x_i, p is piezometric pressure, τ_{ij} is stress tensor components, S_m is mass source, s_i is momentum source components.

For the turbulent flows, u_i,ρ and other dependent variables including τ_{ij} assume their ensemble average values and stress relationship as,

$$\tau_{ij} = 2\mu s_{ij} - \frac{2}{3}\mu \frac{\partial u_k}{\partial x_k}\delta_{ij} - \bar{\rho}\overline{u_i' u_j'} \tag{3}$$

where $u'u'$ is the fluctuating component about the ensemble average velocity and over bar denotes the ensemble average. Right most term in Eq. (3) represents the additional Reynolds stresses due to turbulent motion. These are linked to mean velocity field via turbulence models.

2.3.2 Turbulence Model

In this study, the RNG k-ε turbulence model is used as follows [49].

$$\begin{aligned}\frac{\partial}{\partial t}(\rho k) + \frac{\partial}{\partial x_j}\left(\rho u_j k - \left(\mu + \frac{\mu_t}{\sigma_k}\right)\frac{\partial k}{\partial x_j}\right) \\ = \mu_t(P + P_B) - \rho\varepsilon - \frac{2}{3}\left(\mu_t \frac{\partial u_i}{\partial x_i} + \rho k\right)\frac{\partial u_i}{\partial x_i}\end{aligned} \tag{4}$$

$$\begin{aligned}\frac{\partial}{\partial t}(\rho\varepsilon) + \frac{\partial}{\partial x_j}\left(\rho u_j \varepsilon - \left(\mu + \frac{\mu_t}{\sigma_\varepsilon}\frac{\partial \varepsilon}{\partial x_j}\right)\right) \\ = \mathrm{C}_{\varepsilon 1}\frac{\varepsilon}{k}\left[\mu_t P - \frac{2}{3}\left(\mu_t \frac{\partial u_i}{\partial x_i} + \rho k\right)\frac{\partial u_i}{\partial x_i}\right] + C_{\varepsilon 3}\frac{\varepsilon}{k}\mu_t P_B \\ - C_{\varepsilon 2}\frac{\rho\varepsilon^2}{\mathrm{k}}\mathrm{C}_{\varepsilon 4}\rho\varepsilon\frac{\partial u_i}{\partial x_i} - \frac{\mathrm{C}_\mu \eta^3(1 - \eta/\eta_o)}{1 + \beta\eta^3}\frac{\rho\varepsilon^2}{\mathrm{k}}\end{aligned} \tag{5}$$

The distinctive feature of the RNG k-ε model is the additional last term in the dissipation equation. This arises from the RNG analysis and represents the effect of mean flow distortion on the turbulence. The RNG model coefficients are taken from Rodi et al. [50].

2.3.3 Energy Transport

The compressible form of the energy transport is given by

$$\begin{aligned}\frac{\partial \rho e}{\partial t} + \frac{\partial u_j \rho e}{\partial x_j} = -P\frac{\partial u_j}{\partial x_j} + \sigma_{ij}\frac{\partial u_i}{\partial x_j} + \frac{\partial}{\partial x_j}\left(K\frac{\partial T}{\partial x_j}\right) \\ + \frac{\partial}{\partial x_j}\left(\rho D \sum_m h_m \frac{\partial Y_m}{\partial x_j}\right) + S\end{aligned} \tag{6}$$

where ρ is density, D is the mass diffusion coefficient, P is the pressure, e is the specific internal energy, h_m is the species enthalpy, K is the conductivity, σ_{ij} is the stress tensor, S is the source term, Y_m is the mass fraction of species m and T is temperature. When turbulence model is activated, the conductivity is replaced by the turbulent conductivity, which is given by

$$K_t = K + c_p \frac{\mu_t}{\mathrm{Pr}_t} \tag{7}$$

where Pr_t is the turbulent Prandtl number and μ_t is the turbulent viscosity. The turbulent Prandtl number is given as

$$\mathrm{Pr}_t = \frac{c_p \mu_t}{k_t} \tag{8}$$

where μ_t is the turbulent viscosity, c_p is the specific heat, and k_t is the turbulent conductivity.

In addition to the convection and diffusion terms, the energy equation contains four extra terms. First, a source term is added to account for user-specified energy sources and turbulent dissipation. A pressure work term, $-P\frac{\partial u_j}{\partial x_j}$, accounts for compression and expansion. A viscous dissipation term, $\sigma_{ij}\frac{\partial u_i}{\partial x_j}$, accounts for kinetic energy viscously dissipating into heat. Finally, a species diffusion term, $\frac{\partial}{\partial x_j}\left(\rho D \sum_m h_m \frac{\partial Y_m}{\partial x_j}\right)$, accounts for energy transport due to species diffusion.

2.3.4 Species Transport

The species transport equation solves for the mass fraction of all the species in the domain. The species mass fraction is defined as

$$Y_m = \frac{M_m}{M_{tot}} = \frac{\rho_m}{\rho_{tot}} \tag{9}$$

where M_m is the mass of species m in the cell, M_{tot} is the total mass in the cell, ρ_m is the density of species m, and ρ_{tot} is the density of the cell. The species equations can be solved alone or together with any of the other transport equations. The compressible form of the species conservation equation is given by

$$\frac{\partial \rho_m}{\partial t} + \frac{\partial \rho_m u_j}{\partial x_j} = \frac{\partial}{\partial x_j}\left(\rho D \frac{\partial Y_m}{\partial x_j}\right) + S_m \tag{10}$$

where

$$\rho_m = Y_m \rho \tag{11}$$

and where u is velocity, ρ is density, ρ_m is the species density, Y_m is mass fraction of species m, is the mass diffusion coefficient, and S_m is the source term. The molecular turbulent mass diffusion coefficient is calculated by

$$D_t = \frac{v_t}{Sc_t} \tag{12}$$

where S_c is the Schmidt number.

2.3.5 Spray Break-Up and Combustion Models

The spray break-up is analyzed using the kh-rt model [51] and the SAGE detailed chemistry solver is used for combustion modelling as proposed by Senecal et al. [52].

2.3.6 Solver

In this study point-wise transient successive over relaxation (SOR) algorithm is used to solve the system of algebraic equations that are generated by discretizing the governing equations over the grid. The SOR algorithm is an iterative scheme with a relaxation factor, ω, for accelerating convergence. Given a square system of

$$Ax = B \tag{13}$$

the system can be rewritten as

$$x_i^{k+1} = (1-\omega)x_i^k + \frac{\omega}{a_{ii}}\left(b_i - \sum_{j<1} a_{ij}x_j^{k+1} - \sum_{j>1} a_{ij}x_j^k\right) \tag{14}$$

where $i = 1, 2, \ldots.n$.

In order to aid the convergence the solution is under-relaxed which means that the next iteration is set to a value based on the previous value and a scaled corrector term, as

$$\phi^{**} = \phi^* + \omega(\Delta\phi) \tag{15}$$

where ϕ^{**} is the new iteration value, ϕ^* is the old iteration value, ω is the under-relaxation parameter (usually less than 1.0), and $\Delta\phi$ is the calculated change in the iteration value. The following under-relaxation factors as shown in Table 2 are used for solving various equations.

In order to solve a transient case, the governing equations described earlier are approximated using numerical techniques. The solution is obtained implicitly, which means that an iterative (multi-step) technique will be required for the system of algebraic equations.

A solution is considered converged when the iteration error in the solution is at or below the user-specified convergence criterion. In this study, the iteration error is related to the change in the solution field from each iteration $\Delta\phi$, which is given by

Table 2 Under relaxation factors for different parameters

Equation	Typical value
Momentum	1.0
Pressure	1.3
Density	1.0
Energy	1.0
Species	1.0
TKE	0.7
Epsilon	0.7

$$error = \frac{\Delta\phi}{Norm} \tag{16}$$

where *Norm* is the normalization. Table 3 below summarizes the tolerance parameter and normalization for each equation.

3 Validation of CFD Models

3.1 Validation of Fuel Spray and Break-Up Models

The fuel spray model used is validated using the experimental and the CFD results of Yajia et al. [53] to the extent possible. Yajia et al. [53] investigated the characteristics of the ethanol and gasoline sprays using an eight-hole DISI injector both experimentally and numerically in a cylindrical chamber with the diameter of 50 mm and the height of 100 mm. The temperatures of both ambient air and injected fuel were maintained at 293 K. In all the cases, 15 mg of fuel was injected into the chamber and the fuel injection duration was set to 1.5 ms. They measured the spray penetration lengths of gasoline and ethanol at different combinations of fuel injection and ambient pressures using the experimental setup. However, they presented the spray structure of the ethanol only. In this study, the following parameters are considered for validation: (1) spray penetration lengths of the gasoline sprays with the experimental results of Yajia et al. [53] at ambient pressures of 0.1, 0.45 and 1.0 MPa respectively, and (2) spray structures of the ethanol at 1 ms with the experimental results of Yajia et al. [53]. In all the above cases, the fuel injection pressure is maintained at 5 MPa.

Figure 3 shows the comparison of fuel spray penetration lengths between the experimental results of Yajia et al. [53] and the present CFD analysis, at the ambient pressures of 0.1, 0.45 and 1.0 MPa. From Fig. 3, it is seen that, at all the ambient pressures, the fuel spray penetration lengths predicted by the present CFD analysis are in good agreement with that of the experimental results of Yajia et al. [53].

Table 3 Error tolerance for different equations

Equation	Error tolerance	Normalization
Momentum	1.0e−4	Velocity or 1.0, whichever is larger
Pressure	1.0e−8	Pressure or 1.0, whichever is larger
Density	1.0e−4	Density or 1.0 whichever is larger
Energy	1.0e−4	Internal energy or 1.0, whichever is larger
Species	1.0e−4	1.0
TKE	1.0e−3	Cell TKE
Epsilon	1.0e−3	Cell Epsilon

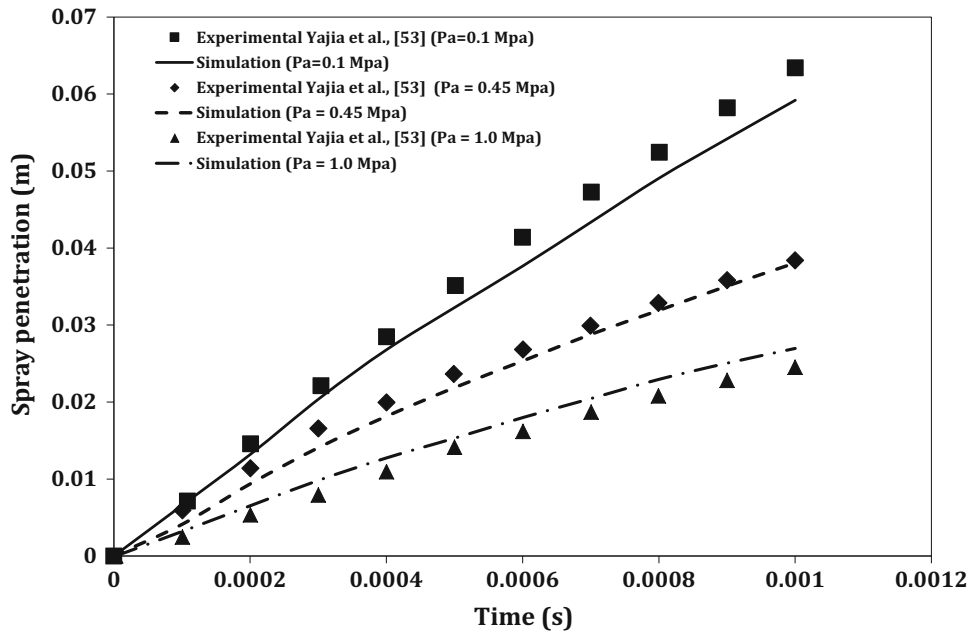


Fig. 3 Comparison of fuel spray penetration lengths at various ambient pressures

Figure 4 shows the comparison of the spray structure of ethanol, at the ambient pressure of 0.1 MPa and the fuel injection pressure of 5 MPa, between the experimental results of Yajia et al. [53] and the present CFD analysis. From Fig. 4, it is seen that there is a good agreement between the two. Therefore, it can be said that the fuel spray structure is well predicted by the present spray breakup model. Thus, it can be used for the further simulation study.

3.2 Validation of Combustion Model

In this study, the combustion model used in the CFD analysis is validated with the available experimental and CFD results of Costa et al. [37], for the engine configuration as mentioned in Table 1. The variation of in-cylinder pressure with the crank angles, the ER distribution on the central vertical plane at 710 CAD and the variation of the carbon dioxide formation with crank angles, obtained by Costa et al. [37] are compared with that of the present CFD study.

Figure 5 shows the comparison of the variation of in-cylinder pressures with crank angles between the experimental results of Costa et al. [37] and the present CFD study. The experimental pressures are the averages of 200 consecutive cycles. From Fig. 5, it is seen that the two results are in good agreement. Therefore, the combustion models used, in this study, can be used for the further analysis with confidence.

Figure 6 shows the comparison of the spatial distribution of ER at the time of ignition (10 CAD before TDC) between the simulation results of Costa et al. [37] and the present CFD analysis, for the fuel injection timing of 300 CAD before TDC. From Fig. 6, it is seen that the ER distribution obtained by the present CFD analysis is very much similar to that obtained by Costa et al. [37].

Figure 7 shows the comparison of the CO_2 formation with crank angles between the simulation results of Costa et al. [37] and the present CFD study. From Fig. 7, it is seen that the two results are in good agreement by both qualitatively and quantitatively.

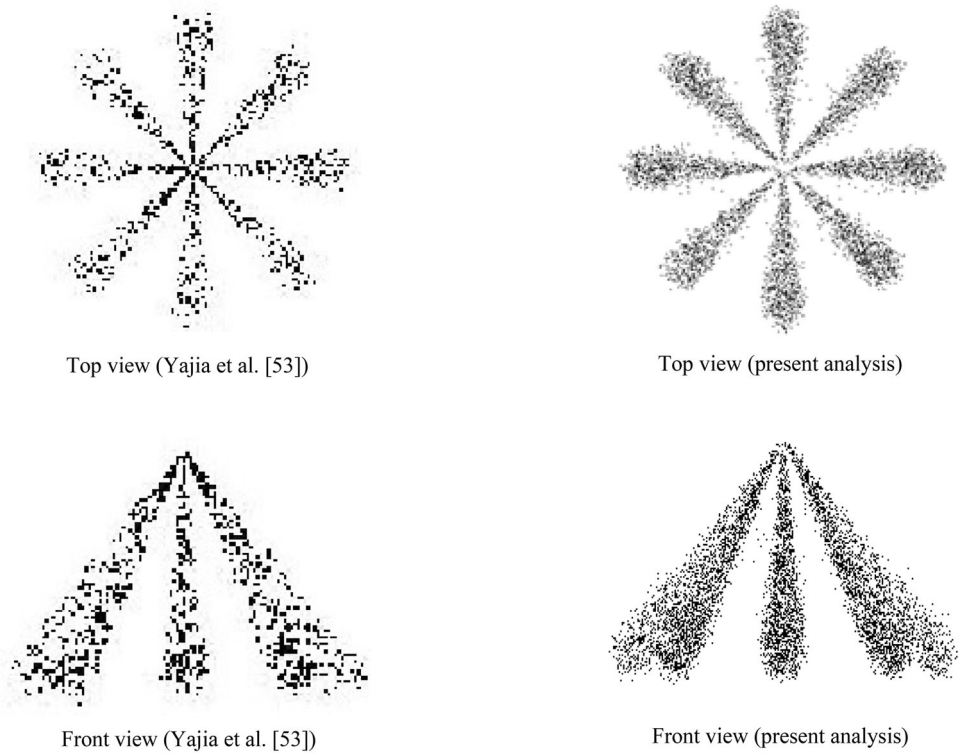


Fig. 4 Comparison of ethanol spray structures

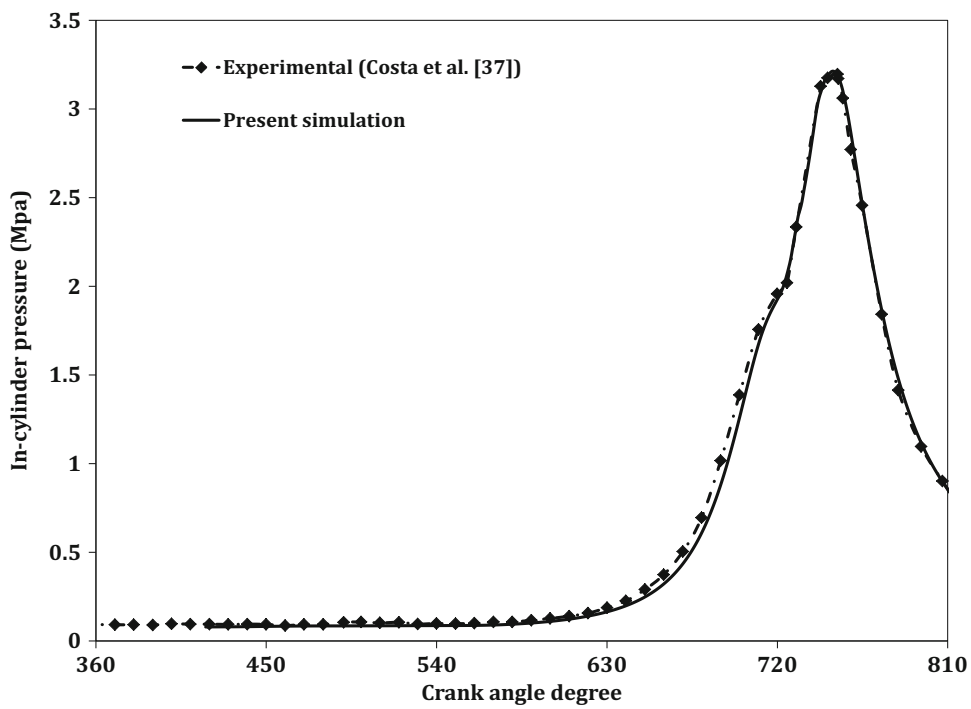


Fig. 5 Comparison of in-cylinder pressures

4 Classification of Mixture Distribution

In this study, the mixture distribution in the combustion chamber is characterized based on the variation of the ER with the distance away from the spark plug location. The combustion chamber is divided into various spherical zones based on their distance from the spark plug as shown in Fig. 8. The spherical zones are defined by considering the spark plug as the center. The radii of the various spherical zones as shown in Fig. 8 are represented by a dimensionless distance parameter '*k*' which is defined as,

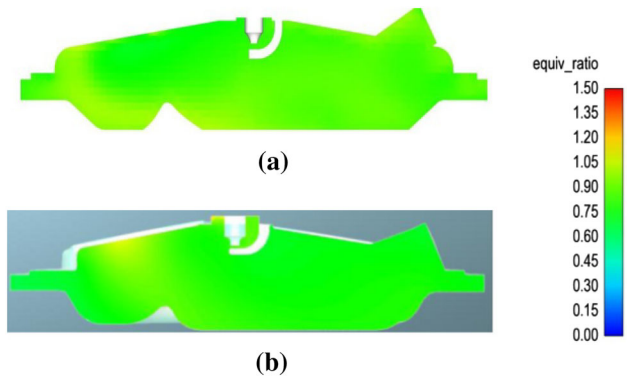


Fig. 6 Comparison of ER distribution on the central vertical plane at the time of spark. **a** From the present CFD analysis. **b** From CFD analysis of Costa et al. [37]

$$k = \frac{r}{R} \tag{17}$$

where R is the radius of the engine cylinder and r is the radial distance between the spark plug and any other point in the combustion chamber. Thus, when r varies between 0 and R, k varies between 0 and 1. Usage of k, instead of discrete zone numbers or the cylinder radius generalizes this procedure for any engine.

The mixture distribution in GDI engines can be broadly categorized into homogeneous and non-homogeneous mixtures.

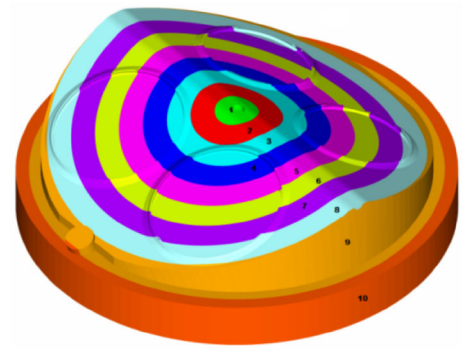


Fig. 8 Creation of spherical zones inside the combustion chamber

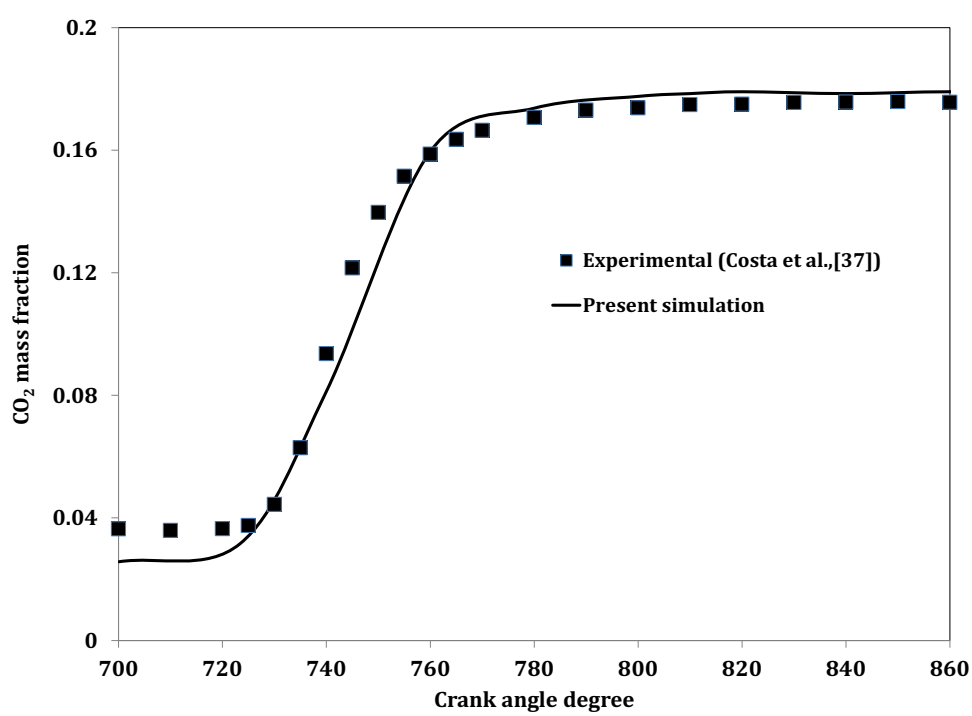
4.1 Homogeneous Mixture

It is a type of mixture distribution in which the spatial variation of the ER with an increase in the distance from the spark plug is almost zero. All other types of mixtures other than this are characterized as *non-homogeneous mixtures*.

4.2 Non-homogeneous Mixture

These can be further classified into properly stratified, improperly stratified and mal-distributed type. According to Spicher et al. [54], in a stratified mixture, the ER should be nearly stoichiometric in the vicinity of the spark plug and it should decrease with the increase in the distance from the spark plug location. Based on this definition, the

Fig. 7 Comparison of CO_2 formation with crank angles



three types of non-homogeneous mixtures are defined as follows.

4.2.1 Properly Stratified Mixture

In this type, the ER at the spark plug location is nearly stoichiometric and it decreases away from the spark plug. This definition does not mention about the rate of change of ER with the distance from the spark plug location. Therefore, four types of variation in the ER in the combustion chamber are considered in this study.

Random Mixture Distribution In this case, there is stoichiometric mixture only around spark plug and the rest of the locations are filled with a lean mixture as shown in Fig. 9a.

4.2.1.1 Linear Mixture Distribution In this case, the variation in the ER with the distance from the spark plug location follows a straight line that has a negative slope while maintaining the stoichiometric mixture at the spark plug location as shown in Fig. 9b.

4.2.1.2 Gaussian Mixture Distribution In this case, the variation in the ER with the distance from the spark plug location follows a Gaussian distribution while maintaining the stoichiometric mixture at the spark plug location as shown in Fig. 9c.

4.2.1.3 Parabolic Mixture Distribution In this case, the variation in the ER with the distance from the spark plug location follows a parabolic distribution while maintaining

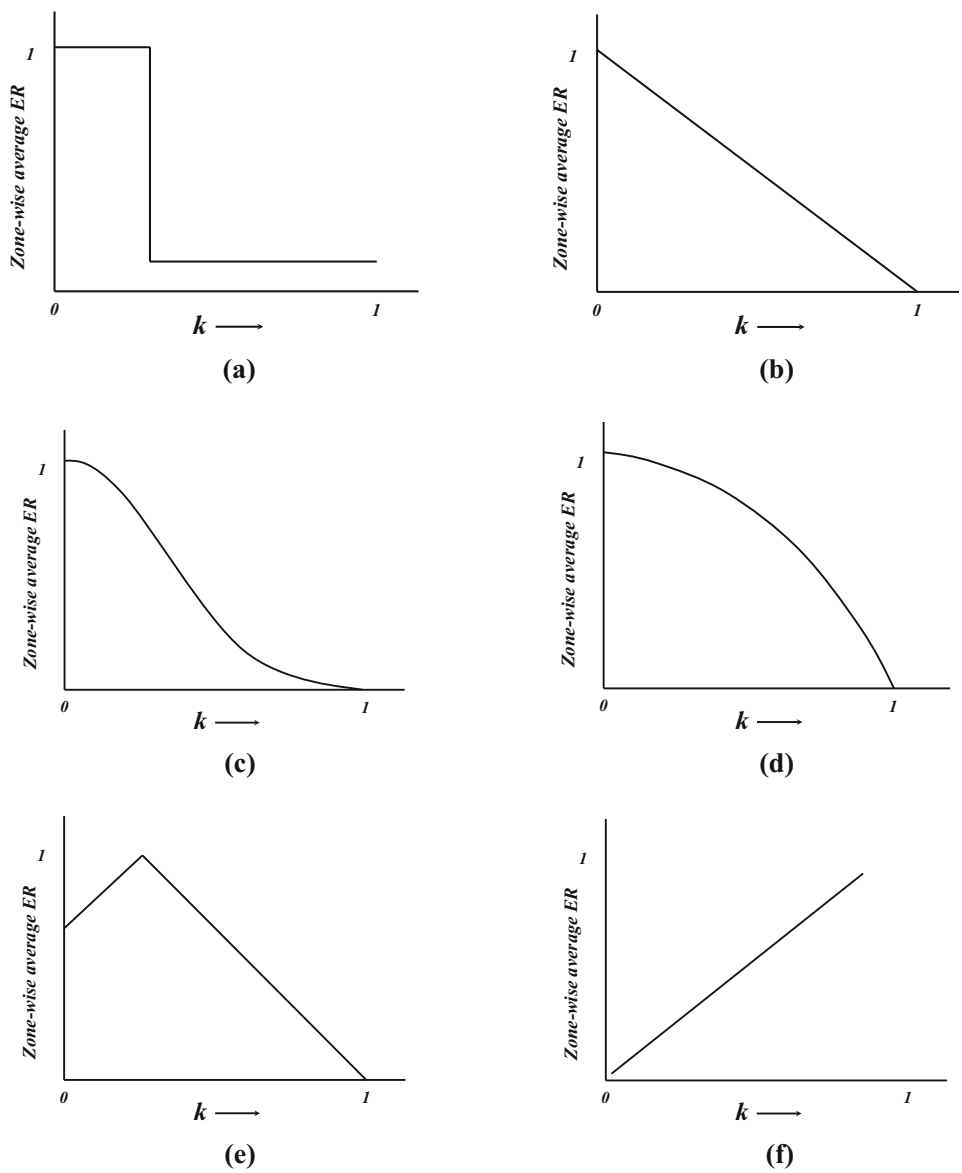


Fig. 9 Variation of zone-wise average ER with k for different mixture distributions. **a** Random variation. **b** Linear variation. **c** Gaussian variation. **d** Parabolic variation. **e** Typical mixture distribution. **f** Maldistribution

the stoichiometric mixture at the spark plug location as shown in Fig. 9).

4.2.2 Improperly Stratified Mixture

In this type, there are combustible or rich mixture zones away from spark plug and the comparatively leaner mixture in the vicinity of the spark plug as shown in Fig. 9e.

4.2.3 Mal-Distributed Mixture

In this type, the ER increases away from the spark plug location as shown in Fig. 9f.

5 Initialization of Species Concentration for Different Cases of Mixture Distribution

In this section, the procedure to initialize the species concentration for different cases of the mixture distributions is discussed. Initially, a CFD simulation is carried out from the exhaust valve opening (EVO) to 1 CAD before the ignition (704 CAD). At the end of this, the cell-wise information of different components of velocity, pressure, temperature, species concentration, TKE and turbulent intensity along with the coordinate position of each cell is extracted. Then, the cell-wise information is sorted and divided into various zones, based on the Euclidian distance from the spark plug. The species concentration values of gasoline, oxygen, and nitrogen, in the extracted data, is replaced with the expected zone-wise values that are separately calculated. The data, thus modified, is saved and used as the initial conditions for the next part of the simulation that starts from 704 CAD up to the exhaust valve opening (EVO). The whole procedure is automated using an in-house Matlab code developed for this purpose. The Matlab code is written in such a way that the overall ER in the combustion chamber is systematically varied by keeping a stoichiometric mixture near the spark plug in all cases and adjusting the zone-wise ER accordingly. Figure 10 shows the mixture distribution for the different variations when the overall ER is 0.5 at the time of ignition.

6 Results and Discussion

This section presents the comparison of the combustion, performance and emission characteristics of the different mixture distribution cases with those of the late fuel injection case (referred to as the baseline case from hereafter) to identify the best mixture distribution among them. The results are presented at various overall ERs ranging from 0.3 to 0.7 in steps of 0.1 (this is the typical range of the overall ER of a GDI engine in the stratified mode of operation). Then, a method is presented to parameterize the mixture distribution in the combustion chamber based on the ideal mixture distribution cases.

Table 4 shows the zone-wise average ER values for the baseline case at different overall ERs. Figure 11 shows the comparison of the variation of zone-wise average ER with k for different mixture distributions at various overall ER. From Fig. 11a, b, it is seen that there is a very lean mixture near the spark plug for the overall ERs of 0.3 and 0.4 respectively. From Fig. 11c, for the overall ER of 0.5, the mixture near the spark plug, for the baseline case, is closer to the stoichiometric value and the average ER gradually decreases away from the spark plug. This is according to the definition of the ideally stratified mixture given in Sect. 4. Also, from Fig. 11d, e, for the overall ER values of 0.6 and 0.7, there is a rich mixture near the spark plug and the average ER decreases away from the spark plug. Thus, it is expected that a poor or no combustion may occur,

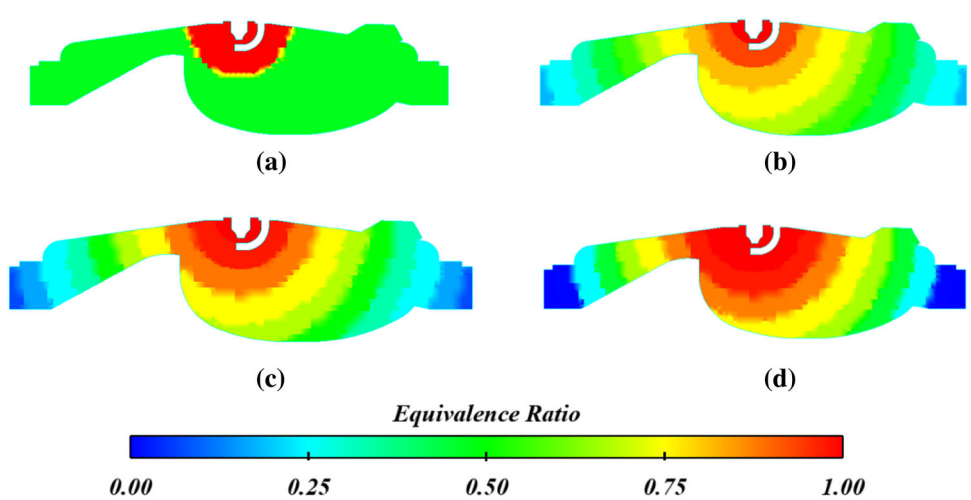


Fig. 10 The ER distribution contours at the time of ignition on the central vertical plane. **a** Random. **b** Linear. **c** Gaussian. **d** Parabolic

Table 4 Zone-wise average ER values of the baseline case for different overall ERs

k	ER = 0.3	ER = 0.4	ER = 0.5	ER = 0.6	ER = 0.7
0.1	0.3	0.24	1.15	1.3	1.6
0.2	0.27	0.24	0.86	1.14	1.21
0.3	0.26	0.4	0.7	1.12	1.05
0.4	0.4	0.58	0.68	0.9	1.04
0.5	0.37	0.55	0.63	0.71	0.92
0.6	0.29	0.4	0.5	0.57	0.74
0.7	0.27	0.36	0.43	0.48	0.61
0.8	0.27	0.35	0.39	0.42	0.48
0.9	0.27	0.31	0.35	0.39	0.41
1	0.24	0.25	0.29	0.31	0.34

when the overall ER is 0.3 and 0.4. A good combustion may occur when the overall ER is 0.5 and a slower or incomplete combustion may occur when the overall ER is 0.6 and 0.7.

6.1 Mixture Distribution on the Central Plane at Different Overall ERs

Figure 12 shows the comparison of the ER distributions, on the central plane of the cylinder, for the baseline case at different overall ERs. From Fig. 12a, b, it is seen that the mixture near the spark plug is not in the combustible range when the overall ER is 0.3 and 0.4. This is because of the combined effect of less fuel quantity injected and an improper fuel injection timing, however, from Fig. 12c–e, it is seen that there is a combustible/rich mixture near the spark plug, when the overall ER is 0.5, 0.6 and 0.7 respectively.

6.2 Effect of Mixture Distribution on the In-Cylinder Pressure

Figure 13 shows a comparison of the variations of in-cylinder pressures with crank angles for different mixture distribution cases. Table 5 summarizes the percentage variation of the peak in-cylinder pressure with respect to that of the baseline case.

From Fig. 13a, b, when the overall ER is 0.3 and 0.4, it is seen that the in-cylinder pressure curve of the baseline case follows the motoring curve which implies no combustion. This is because of the absence of the combustible mixture near the spark plug at the time of ignition. From Fig. 13 and Table 5, it is seen that, the parabolic mixture distribution results in a higher in-cylinder pressure than that of the other cases at all the overall ERs. At the overall ERs of 0.5, 0.6 and 0.7, the random mixture distribution gives lower peak in-cylinder pressure compared to that of the baseline case. Whereas at the overall ER of 0.6, the peak in-cylinder pressure of the base case matches with that of the linear mixture distribution case. However, under all other conditions, the peak in-cylinder pressures, for the base case, is lower than that of the other cases as shown in Table 5.

From the above discussion, it can be concluded that the mixture distribution based on the parabolic case is desirable in order to achieve the highest peak in-cylinder pressure at all the overall ERs.

6.3 Effect of Mixture Distribution on the In-Cylinder Temperature

Figure 14 shows a comparison of the variations of in-cylinder temperatures with crank angles for different mixture distributions. Table 6 shows the percentage variations of the peak in-cylinder temperature with respect to the baseline case. From Fig. 14a, b, it is seen that the peak in-cylinder temperature is lower for the baseline case, compared to that of the other mixture distribution cases. This is because of no combustion in these cases that resulted from a lack of combustible mixture near the spark plug at the time of ignition as discussed in Sect. 6.1. From Fig. 14, it is seen that the peak in-cylinder temperature, at all the overall ERs, is higher in the case of parabolic mixture distribution case compared to that of the other cases. From Fig. 14c and Table 6, when the overall ER is 0.5, it is seen that the peak in-cylinder temperature of the baseline case is similar to that of the linear mixture distribution case. Similarly, from Fig. 14d, it is seen that the peak in-cylinder temperature of the baseline case is similar to that of the Gaussian mixture distribution case. From Fig. 14e, when the overall ER is 0.7, the peak in-cylinder temperature for the baseline case is similar to that of the random mixture distribution case with the only variation in the time of its occurrence.

From the above discussion, it can be concluded that the peak in-cylinder temperature, at all the considered overall ERs, occurs for the parabolic mixture distribution case, followed by that of the Gaussian, linear and random mixture distribution cases. Also, the peak in-cylinder temperature of the baseline case approaches that of the Gaussian or the parabolic cases when the stratification index is closer to 1.

6.4 Effect of Mixture Distribution on the Heat Release Rate

Figure 15 shows the comparison of the variations of heat release rates with crank angles for various mixture

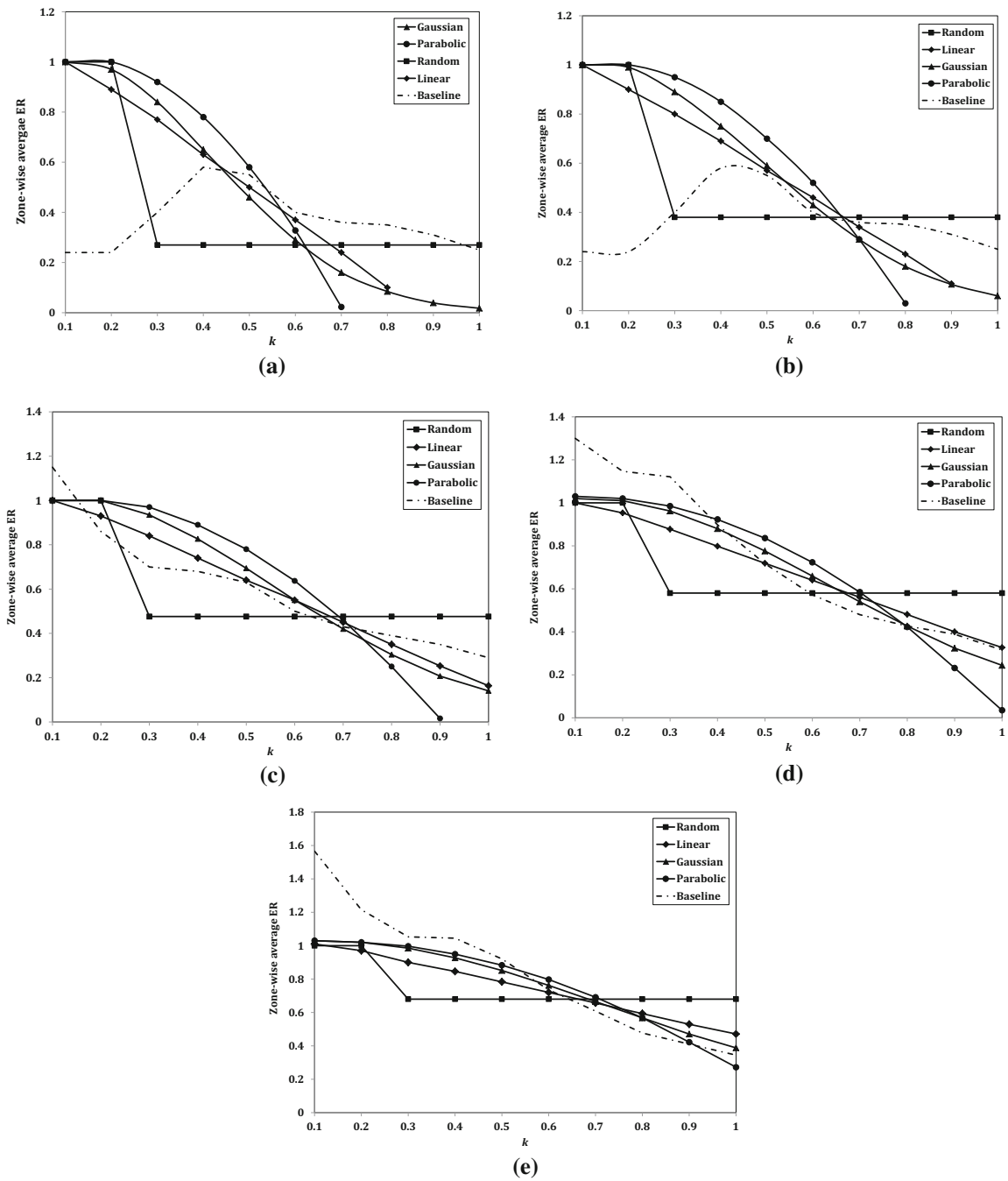


Fig. 11 Variation of the zone-wise average ER with k for different mixture distributions. **a** ER = 0.3. **b** ER = 0.4. **c** ER = 0.5. **d** ER = 0.6. **e** ER = 0.7

distribution cases. Table 7 shows the percentage variation of the peak heat release rate with respect to that of the baseline case.

In Table 7, the percentage variations of heat release rates for different mixture distribution cases with respect to that of the baseline case for the overall ERs of 0.3 and 0.4 is not shown because of any combustion. This can be seen from Fig. 15a, b. However, at the overall ERs of 0.5, 0.6 and 0.7, the random mixture distribution case has significantly lower heat release rate compared to that of the baseline case. At all the other conditions, the heat release rate of the baseline case is lower than that of the other ideal mixture distribution cases.

It is concluded that the mixture distribution based on the parabolic distribution offers better heat release characteristics compared to that of the other mixture distribution cases. Also, the mixture distribution in the baseline case

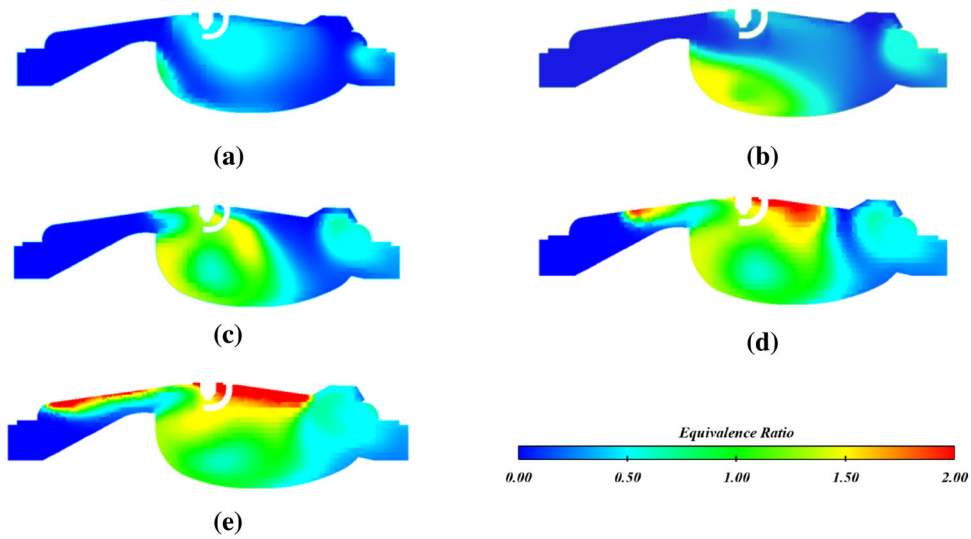


Fig. 12 Comparison of the ER distributions at the time of spark for the baseline case at different ERs. **a** Overall ER = 0.3. **b** Overall ER = 0.4. **c** Overall ER = 0.5. **d** Overall ER = 0.6. **e** Overall ER = 0.7

can be improved at par with the parabolic mixture distribution by undertaking full-fledged optimization study.

6.5 Effect of Mixture Distribution on the IMEP

Figure 16 shows the comparison of IMEP for different mixture distribution cases at various overall ERs. Table 8 gives the summary of the percentage variation of the IMEP for different mixture distribution cases with respect to the baseline case.

From Fig. 16, it is seen that, at the overall ER of 0.3 and 0.4, the IMEP of the baseline case is negligible because of no combustion. Therefore, the percentage variation of the IMEP is not shown in Table 8 for the overall ER of 0.3 and 0.4. However, at the overall ER of 0.5, 0.6 and 0.7, the IMEP of the baseline case is comparable with that of the other mixture distribution cases. From Fig. 16, it is seen that at the overall ER of 0.5, 0.6 and 0.7, the IMEP of the baseline case lies in between that of the random and the linear mixture distribution cases.

From Table 8, it is seen that the parabolic mixture distribution case results in the higher IMEP compared to that of all the other mixture distribution cases. Also, the IMEP of the random mixture distribution case is lower than that of the baseline case, at the overall ER of 0.5, 0.6 and 0.7. The IMEP of linear and Gaussian mixture distribution cases lie in between the baseline and parabolic mixture distribution cases.

6.6 Effect of Mixture Distribution on NO_x Emissions

Figure 17 shows the comparison of NO_x emissions of the baseline case with the other mixture distribution cases, for different overall ERs, at the EVO. From Fig. 17, it is seen that, at all the overall ERs, the NO_x emissions are the highest for the parabolic mixture distribution case. This is because of the nature of mixture distribution that led to higher in-cylinder temperatures compared to the other cases. From Fig. 17, it is also seen that, when the overall ER is 0.3 and 0.4, the NO_x emissions for the baseline and the random mixture distribution cases are zero, which is because of lower peak in-cylinder temperature during combustion. At the overall ER of 0.5, 0.6 and 0.7, the NO_x emissions of the baseline case lie in between those of the random and the linear mixture distribution cases. Table 9 summarizes the percentage variation of NO_x emissions at the EVO for different mixture distribution cases with respect to the baseline case.

6.7 Effect of Mixture Distribution on the HC Emissions

Figure 18 shows the comparison of the HC emissions for various mixture distribution cases, at different overall ERs, at the EVO with that of the baseline case. At all the considered overall ERs, the HC emissions are the highest for the baseline case, and the lowest for the parabolic case. Table 10 shows the percentage variation of HC emissions for the different mixture distribution cases with respect to the baseline case at the EVO. From Fig. 18, it is seen that at the overall ER of 0.7, the HC emissions of all the cases except for the baseline case are negligible. Therefore, the percentage variation of HC emissions for the different mixture distribution cases with respect to the baseline case, at the overall ER of 0.7 is not presented.

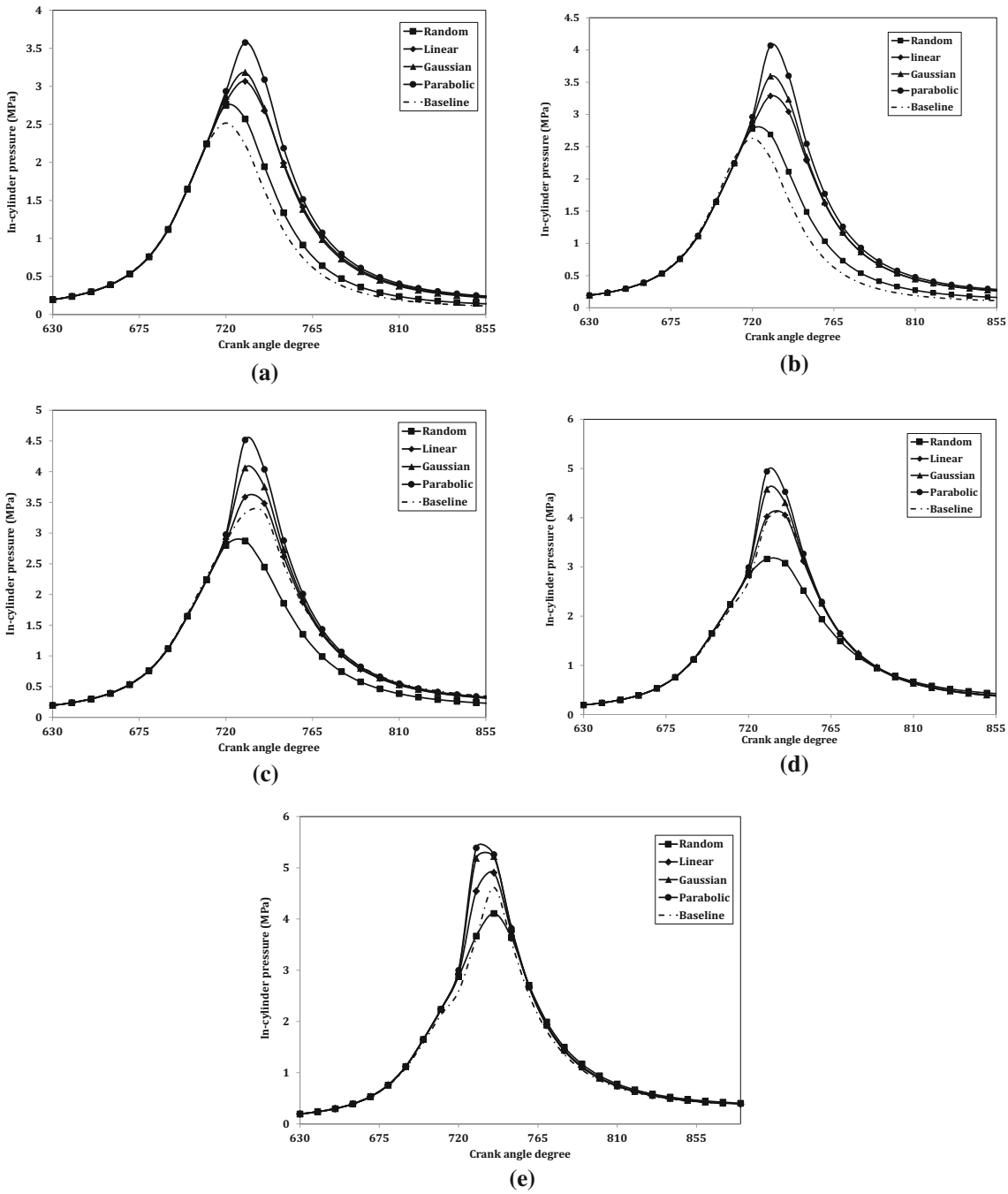


Fig. 13 Comparison of the in-cylinder pressures at different mixture distribution cases. **a** ER = 0.3. **b** ER = 0.4. **c** ER = 0.5. **d** ER = 0.6. **e** ER = 0.7

6.8 Mixture Parametrization Based on Various Mixture Distribution Cases

From the above discussion, it is found that the combustion and performance of a GDI engine depend not only on the ER in the vicinity of the spark plug but also on the distribution in the entire combustion chamber. It is also concluded that the parabolic mixture distribution results in the best combustion and performance characteristics in the GDI engine. This section presents the parametrization of in-cylinder mixture based on the various mixture distribution cases discussed earlier. It involves, creating some mathematical parameters for identifying the rich mixture zones and their location in the combustion chamber and ensuring a mixture that gradually becomes leaner with the distance from the spark plug location. These mathematical

Table 5 Percentage variation of in-cylinder peak pressure with respect to the base line case

ER	0.3	0.4	0.5	0.6	0.7
Random	+ 11	+ 13	− 16	− 25	− 30
Linear	+ 22	+ 33	+ 6	0	+ 8
Gaussian	+ 27	+ 45	+ 9	+ 11	+ 20
Parabolic	+ 44	+ 65	+ 32	+ 20	+ 24

parameters can be further used to optimize the mixture distribution in the combustion chamber. In this section, the mixture distributions based on the linear, Gaussian and parabolic cases are considered for parametrization. Parametrization of random mixture distribution is not undertaken as it is not possible to represent it mathematically.

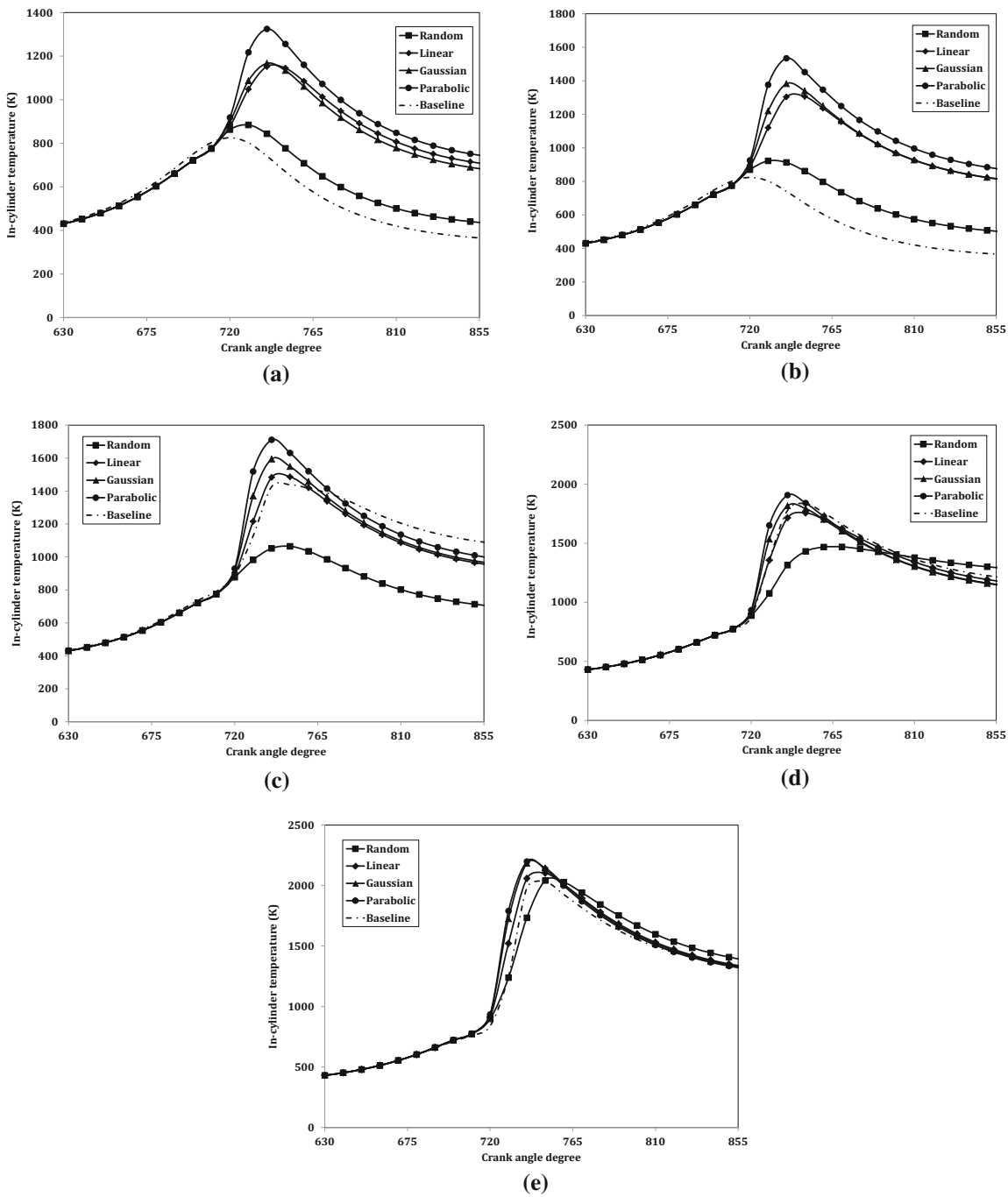


Fig. 14 Comparison of the in-cylinder temperatures for different mixture distribution cases. **a** ER = 0.3. **b** ER = 0.4. **c** ER = 0.5. **d** ER = 0.6. **e** ER = 0.7

Table 6 Percentage variation of peak in-cylinder temperature with respect to the base line case

	0.3	0.4	0.5	0.6	0.7
Random	+ 7	+ 12	− 26	− 20	0
Linear	+ 40	+ 60	+ 2	− 5	+ 3
Gaussian	+ 41	+ 68	+ 10	+ 1.3	+ 6
Parabolic	+ 60	+ 86	+ 18	+ 3	+ 7

6.8.1 Parametrization of Linear Mixture Distribution

In the linear mixture distribution case, the mixture stratification is parametrized by the slope of the straight line. Let the equation of the straight line that represents the variation of ER with the distance from spark plug as

$$y = -m_i x + c \tag{18}$$

The negative value of m_i represents the negative slope of the line. For any general case, initially, the variation in the ER with the distance from the spark plug is plotted. Then, a linear trend line is fitted to represent this variation. Let the equation of the linear trend line be,

$$y = -m_a x + c \tag{19}$$

where m_a represents the slope of the linear trend line of the actual case.

Now, the stratification index with the linear mixture distribution (SI_l) is defined as,

$$SI_l = \frac{-m_a}{-m_i} \tag{20}$$

If the SI_l is equal to 1, the mixture distribution, in the actual case, is same as the ideally stratified case. If the SI_l is less than 1, the mixture distribution is improperly stratified. If the SI_l is more than 1, there are excessively rich mixture zones at the spark plug location. In other words, the value of the SI_l closer to zero indicates poor mixture stratification. If the SI_l is negative, the mixture is mal-distributed. Also, the R-square value of the linear fit obtained for the actual case gives us some idea into the mixture distribution trends in the other zones of the combustion chamber. A lower R-square value implies a mixture distribution that is not similar to the linear mixture distribution case and an R-square value closer to 1 implies a mixture that is closer to the ideal one.

Figure 19 shows the comparison of the linear trend lines of the mixture distribution for the baseline case with the ideal mixture with linear distribution at different overall ERs. Table 11 shows the comparison between the equations of the linear trend lines for the baseline case and the ideal mixture with a linear variation at different overall ERs.

In Fig. 19, the lines that have same overall ER are represented with the same colour for easy comparison. From Fig. 19 and Table 11, it is seen that the linear trend line for the baseline case is closely matching its ideal counterpart at the overall ER of 0.5, which also means that the SI_l of the baseline case is closer to 1. Therefore, it is found that the IMEP (Fig. 16) of the baseline case, at the overall ER of 0.5 is very close to that of the linear mixture distribution case. It is also to be noted that the IMEP of the baseline case, at the other overall ERs, is considerably lower compared to that of its ideal linear variation counterpart.

This is because, at all the other overall ERs, other than 0.5, the SI_l is either far greater or far lesser than 1. Also from Table 11, it is seen that the R-square value is closer to 1 for the overall ER of 0.5, 0.6 and 0.7, whereas it is closer to 0 for the overall ERs of 0.3 and 0.4. This implies that the mixture distribution across the entire combustion chamber is far from the linear trend at the overall ERs of 0.3 and 0.4, whereas it is very close to the linear trend for the other overall ERs.

The limitation of this method is that there is no way to identify the location of the rich zone if it is away from the spark plug. Also, two lines with the same slope could represent different variation in mixture distribution away from the spark plug. While setting up an optimization case, it is required to define the objective function that minimizes the $abs(SI_l - 1)$ with a constraint on R-square to be greater than 0.9.

6.8.2 Parametrization of Gaussian Mixture Distribution

In the Gaussian mixture distribution case, the mixture stratification is parametrized by the peak of the Gaussian distribution and its position. Let the equation of the Gaussian mixture distribution for the ideal case be,

$$y = a_i \exp\left[-\left(\frac{x - b_i}{c_i}\right)^2\right] \tag{21}$$

where a_i represents the peak value of the Gaussian variation, b_i represents the location of the peak value on x-axis and c_i is a constant.

For any general case, initially, the variation in the ER away from the spark plug is plotted. Then a Gaussian trend line is fitted to represent this variation. Let the equation of the Gaussian trend line be,

$$y = a_a \exp\left[-\left(\frac{x - b_a}{c_a}\right)^2\right] \tag{22}$$

where a_a represents the peak value of the Gaussian fit, b_a represents the location of the peak value on the x-axis and c_a is a constant. In order to get meaningful results, a_a and

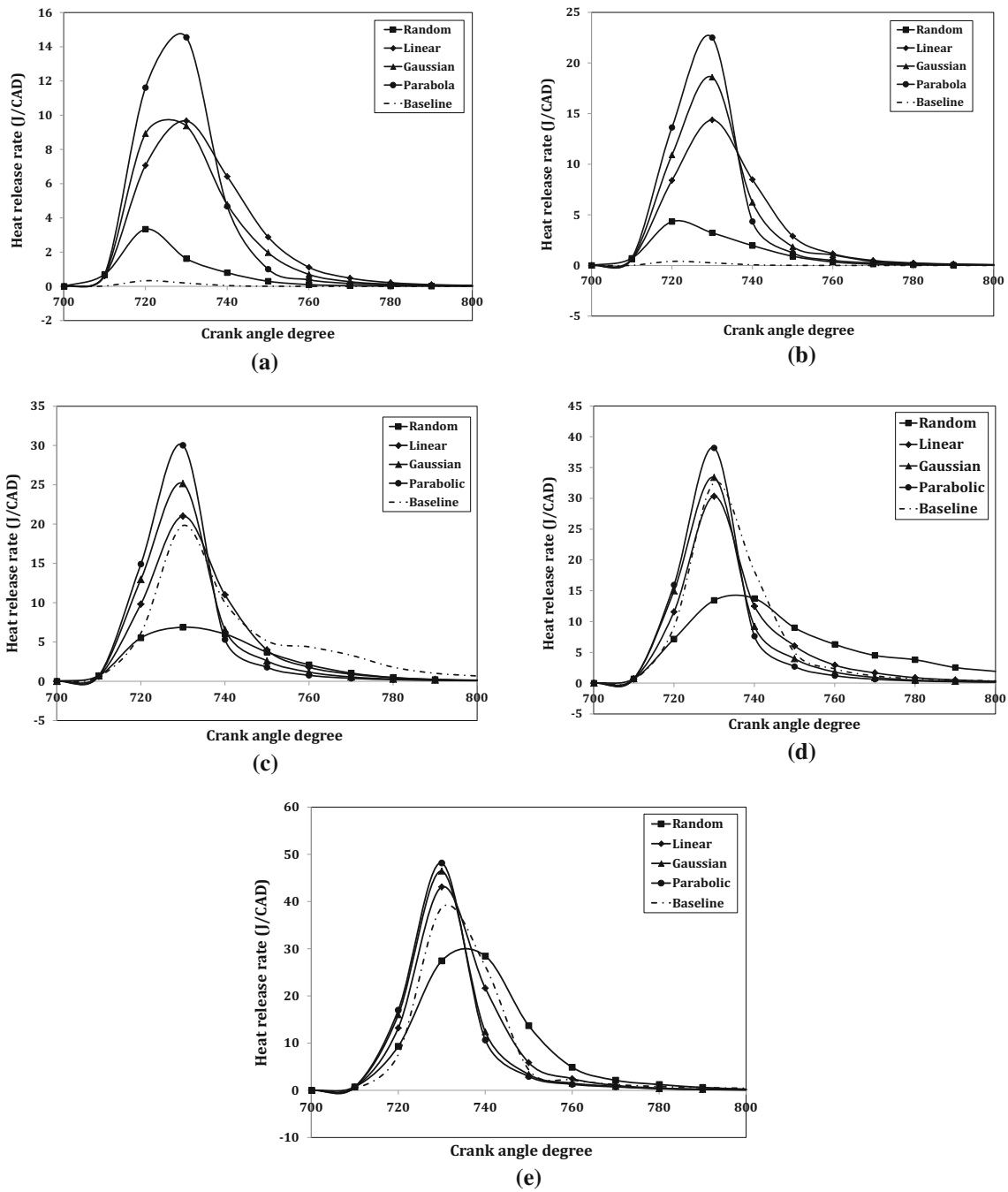


Fig. 15 Comparison of the heat release rate for different mixture distribution cases. **a** ER = 0.3. **b** ER = 0.4. **c** ER = 0.5. **d** ER = 0.6. **e** ER = 0.7

Table 7 Percentage variation of peak heat release rate with respect to the base line case

	0.3	0.4	0.5	0.6	0.7
Random	–	–	− 65	− 57	− 26
Linear	–	–	+ 6	− 6	+ 11
Gaussian	–	–	+ 27	+ 2	+ 20
Parabolic	–	–	+ 51	+ 17	+ 24

b_a values used in this study are bound by the range (0, 2) and (0, 1) respectively. Now, the stratification index with a Gaussian mixture distribution (SI_g) is defined as,

$$SI_g = \frac{-a_a}{-a_i} \tag{23}$$

For an ideally stratified mixture, b_a should be zero. If $b_a \approx 0$ and $SI_g = 1$, then the mixture is ideally stratified. If $b_a \approx 0$ and $SI_g > 1$, then the mixture near the spark plug is

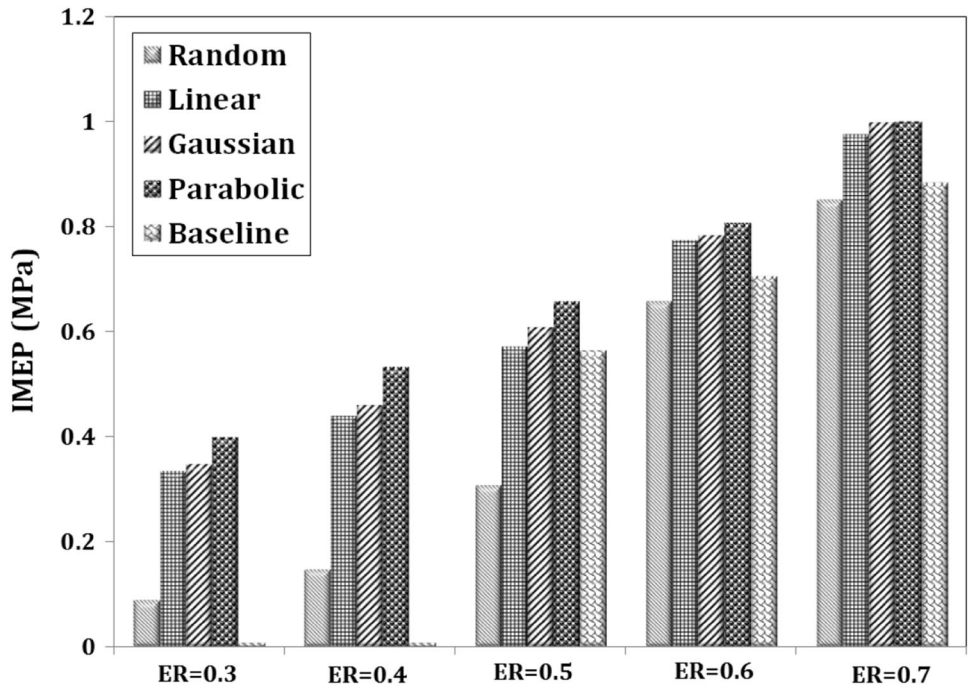


Fig. 16 Comparison of the IMEP for different mixture distribution cases

Table 8 Percentage variation of IMEP with respect to the baseline case

	0.3	0.4	0.5	0.6	0.7
Random	–	–	− 45	− 6	− 3.5
Linear	–	–	+ 1.3	+ 9	+ 11
Gaussian	–	–	+ 7.8	+ 10	+ 13
Parabolic	–	–	+ 16.7	+ 14	+ 13.2

excessively rich. If $0 \ll b_a < 1$, then the mixture is improperly stratified with a peak ER value approximately equal to SI_g at $k = b_a$. If $b_a \approx 1$, then the mixture is maldistributed. Also, the R-square value of the Gaussian fit represents the proximity of the zone-wise ER to the corresponding ideal Gaussian mixture distribution. A lower R-square value implies a mixture distribution that is not similar to the Gaussian mixture distribution case and an

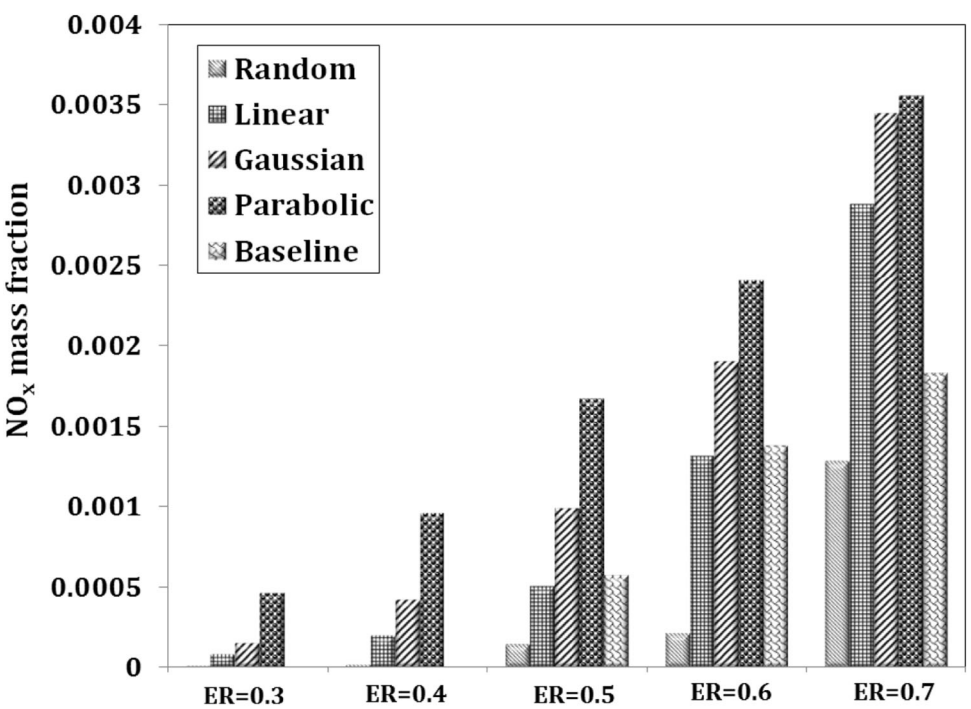


Fig. 17 Comparison of the NO_x emissions for different mixture distribution cases

Table 9 Percentage variation of NO_x at EVO with respect to the baseline case

	0.3	0.4	0.5	0.6	0.7
Random	–	–	− 75	− 85	− 30
Linear	–	–	+ 1	+ 4.8	+ 58
Gaussian	–	–	+ 72	+ 38	+ 89
Parabolic	–	–	+ 190	+ 74	+ 95

Table 10 Percentage variation of HC emissions at the EVO with respect to the baseline case

	0.3	0.4	0.5	0.6	0.7
Random	− 24	− 31	− 11.5	− 8.6	–
Linear	− 75	− 79	− 71	− 23	–
Gaussian	− 77	− 81	− 77	− 27	–
Parabolic	− 91	− 92	− 90	− 52	–

R-square value that is closer to 1 implies a mixture that is closer to the ideal one. While setting an optimization case, the objective function should be defined to minimize the absolute value of b_a, and the $abs(SI_g-1)$ while constraining R-square t be greater than 0.9. This ensures that the stoichiometric mixture remains closer to the spark plug and the mixture gets progressively leaner away from the spark plug following a Gaussian trend.

Figure 20 shows the comparison of the Gaussian trend lines of the mixture distribution for the baseline case, with that of the ideal Gaussian mixture distribution case, at different overall ERs. Table 12 shows the comparison between the equations of the Gaussian trend lines for the baseline case and the ideal Gaussian mixture distribution case at different overall ERs. In Fig. 20, the lines that have same overall ER are represented with the same colour for easy comparison. From Fig. 20 and Table 12, it is seen that the b_a is equal 0 for the overall ERs of 0.5, 0.6 and 0.7 which implies that the richest mixture in the combustion chamber is in the vicinity of spark plug for these cases. However, the SI_g for the overall ER of 0.5 is closer to 1, whereas it is greater than 1 for the overall ERs of 0.6 and 0.7 which implies that the maximum ER in the combustion chamber is closer to 1 (stoichiometric) for the overall ER of 0.5 and it is more than 1 (rich mixture) in the cases when the overall ER is 0.6 and 0.7. Although the b_a is equal to 0 and the SI_g is closer to 1, for the baseline case, at the overall ER of 0.5, R-square value is about 0.85 which implies that the mixture distribution is not as close to the ideal Gaussian mixture distribution as it is for the linear case (R-square value is 0.91, Table 11). Therefore, it is seen that the IMEP of the baseline case is closer to the linear mixture distribution case compared to that of the Gaussian mixture distribution case (Fig. 16). It is also seen that the IMEP of the baseline case, at the other overall ERs, is considerably lower compared to that of the Gaussian mixture distribution case. This is because, at all the other overall ERs, other than 0.5, either the SI_g and the R-square value are not closer to 1 or the b_a is not closer to 0.

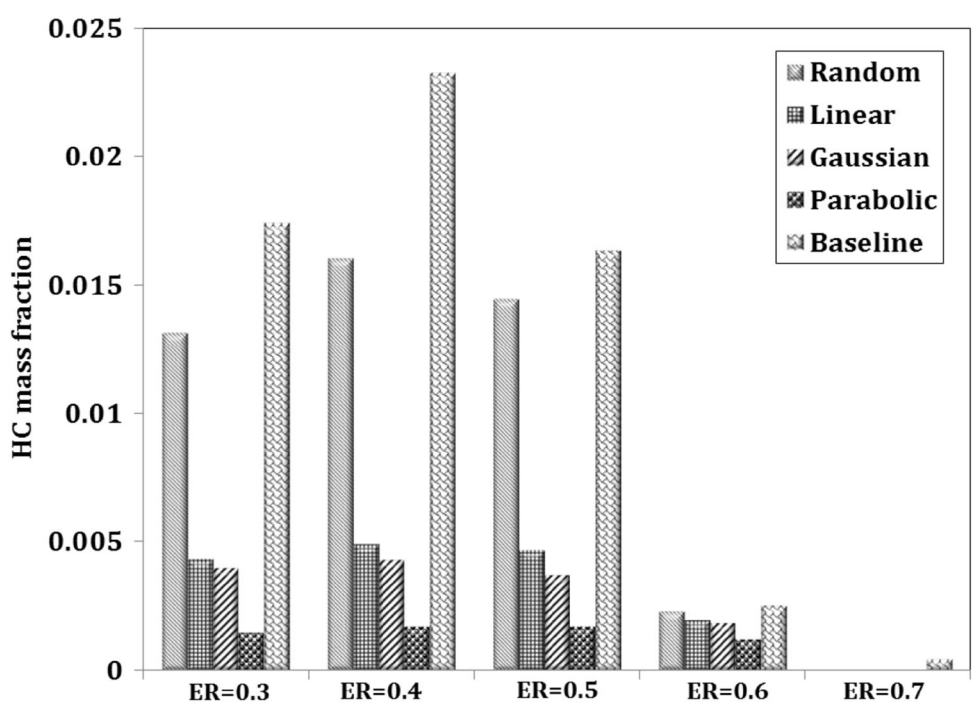


Fig. 18 Comparison of the HC emissions for the different mixture distribution cases

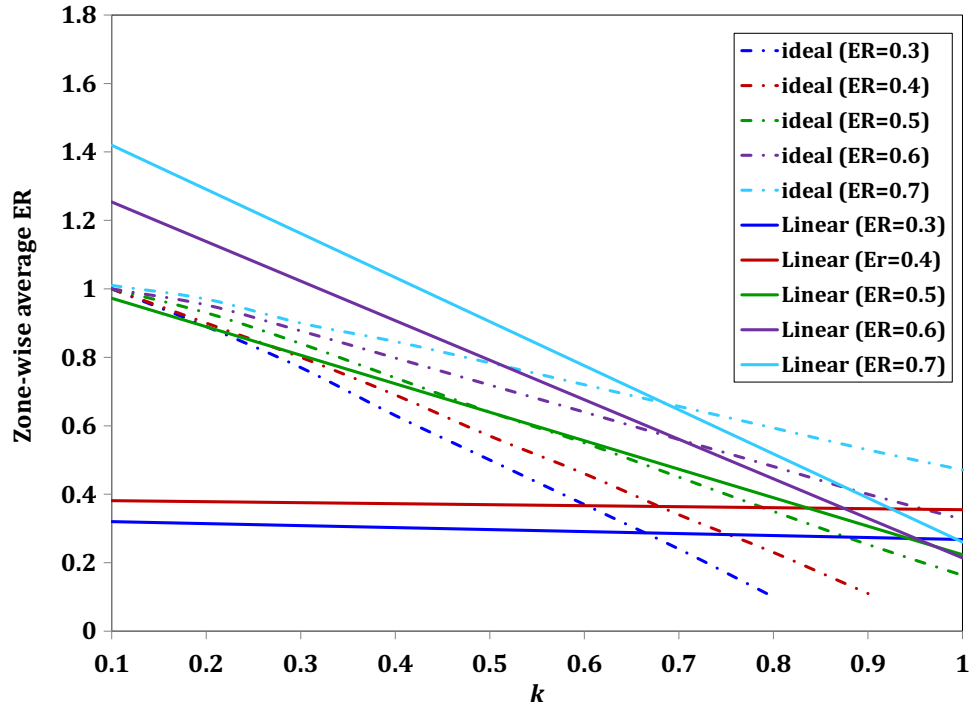


Fig. 19 Comparison of linear fits of the baseline case with that of the linear mixture distribution case at different ERs

Table 11 Comparison of the linear fit equations of the baseline and ideal cases at different ERs

ER	Baseline linear fit	Ideal case	SI_l	R-square (baseline case)
0.3	$y = -0.0291x + 0.384$	$y = -1.29x + 1.1454$	0.022	0.1191
0.4	$y = -0.0582x + 0.326$	$y = -1.12x + 1.1267$	0.052	0.0053
0.5	$y = -0.9327x + 1.056$	$y = -0.9503x + 1.1143$	0.981	0.9096
0.6	$y = -1.1552x + 1.3693$	$y = -0.7698x + 1.0989$	1.5	0.9569
0.7	$y = -1.2885x + 1.5487$	$y = -0.6119x + 1.0846$	2.10	0.9562

6.8.3 Parametrization of Parabolic Mixture Distribution

In the parabolic mixture distribution case, the mixture stratification is parametrized by the coordinates of the vertex of the parabolic distribution. Let the equation of the parabolic mixture distribution for the ideal case be,

$$(x - a_i)^2 = -4(y - b_i) \tag{24}$$

where (a_i, b_i) is the vertex of the parabola. For an ideally stratified case, $(a_i, b_i) = (0, 1)$.

The negative sign on the LHS of Eq. (13) indicates a parabola that is open downwards. For any general case, initially, the variation in the ER away from the spark plug location is plotted. Then, a parabolic trend line is fitted to represent this variation. After re-arranging the equation, let the equation of the fit be,

$$(x - a_a)^2 = -4(y - b_a) \tag{25}$$

where (a_a, b_a) is the vertex of the parabolic fit.

Now, the stratification index with a parabolic mixture distribution (SI_p) is defined as,

$$SI_p = \frac{-b_a}{-b_i} \tag{26}$$

For an ideally stratified mixture, the a_a should be zero. If $a_a \approx 0$ and $SI_p = 1$, then the mixture is ideally stratified. Note that the mixture characterization based on the parabolic variation holds good in identifying the rich mixture zones if and only if the LHS of the Eq. (13) is negative. If it is positive, then the vertex of the parabola represents the leanest mixture and its location in the combustion chamber. Thus, this method is limited in its use.

Figure 21 shows the comparison of the parabolic trend lines of the mixture distribution of the baseline case with the ideal parabolic mixture distribution case at different overall ERs. Table 13 shows the comparison between the equations of the parabolic trend lines for the baseline case and the parabolic mixture variation case at different overall ERs. In Fig. 21, the lines that have same overall ER are represented with the same colour. From Fig. 21 and Table 13, it is seen that the polynomial trend lines, for the baseline case, at the ERs of 0.5, 0.6 and 0.7 are upward

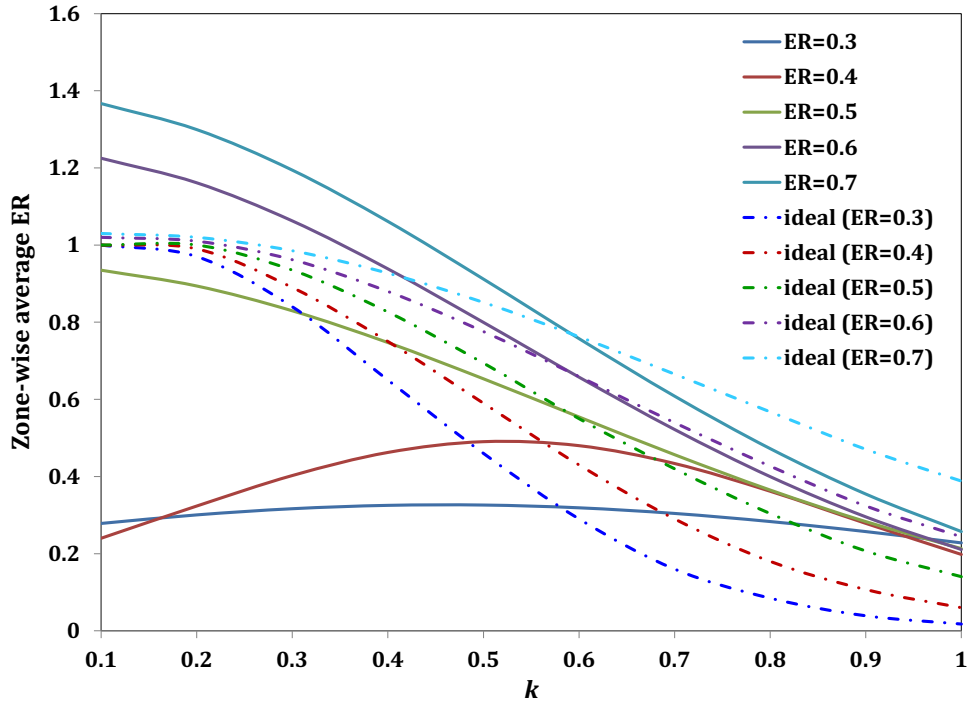


Fig. 20 Comparison of Gaussian fits of the baseline case at different ERs

Table 12 Comparison of the Gaussian fit equations of the baseline and ideal cases at different ERs

ER	Baseline Gaussian fit	Ideal Gaussian equation	SI_g	b_a	R-square
0.3	$y = 0.3267 * \exp\left[-\left(\frac{x-0.4601}{0.9}\right)^2\right]$	$y = 1 * \exp\left[-\left(\frac{x-0}{0.44}\right)^2\right]$	0.3267	0.46	0.2
0.4	$y = 0.4914 * \exp\left[-\left(\frac{x-0.5233}{0.5}\right)^2\right]$	$y = 1 * \exp\left[-\left(\frac{x-0}{0.53}\right)^2\right]$	0.4914	0.523	0.6
0.5	$y = 0.9489 * \exp\left[-\left(\frac{x-0}{0.8184}\right)^2\right]$	$y = 1 * \exp\left[-\left(\frac{x-0}{0.63}\right)^2\right]$	0.95	0	0.85
0.6	$y = 1.247 * \exp\left[-\left(\frac{x-0}{0.75}\right)^2\right]$	$y = 1 * \exp\left[-\left(\frac{x-0}{0.741}\right)^2\right]$	1.25	0	0.96
0.7	$y = 1.39 * \exp\left[-\left(\frac{x-0}{0.77}\right)^2\right]$	$y = 1 * \exp\left[-\left(\frac{x-0}{0.9}\right)^2\right]$	1.39	0	0.93

opening parabolas. Therefore, the a_a for these cases represents the location of the leanest mixture in the combustion chamber instead of the richest one.

When the overall ER is 0.3, the parabolic trend line is downward opening, with the vertex at (0.46, 0.33). This implies that the richest zone in the combustion chamber is located at the k value of 0.46 with the average ER of 0.33. Similarly, when the overall ER is 0.4, the vertex of the parabola is at (0.53, 0.47) which implies that the richest zone in the combustion chamber is at the k value of 0.53 with an average ER of 0.47. Because of its limitation in identifying the location of rich mixture zones, the mixture parametrization based on the parabolic variation is limited in its use.

7 Conclusions

This paper deals with identifying the mixture distribution that gives highest combustion and performance characteristics at a given overall ER in a GDI engine operating under stratified conditions. The GDI engine operating with the late fuel injection is considered to be the baseline case. Four types of mixture distributions viz., random, linear, Gaussian and parabolic, are studied and their combustion, performance, and emission characteristics are compared with that of the baseline case at different overall ERs. Then the mixture distribution is parametrized into mathematical equations in order to define some objective functions that can be used for optimization of the mixture distribution. The following conclusions are drawn:

- At all the overall ERs, the parabolic mixture distribution case gives the best combustion and performance

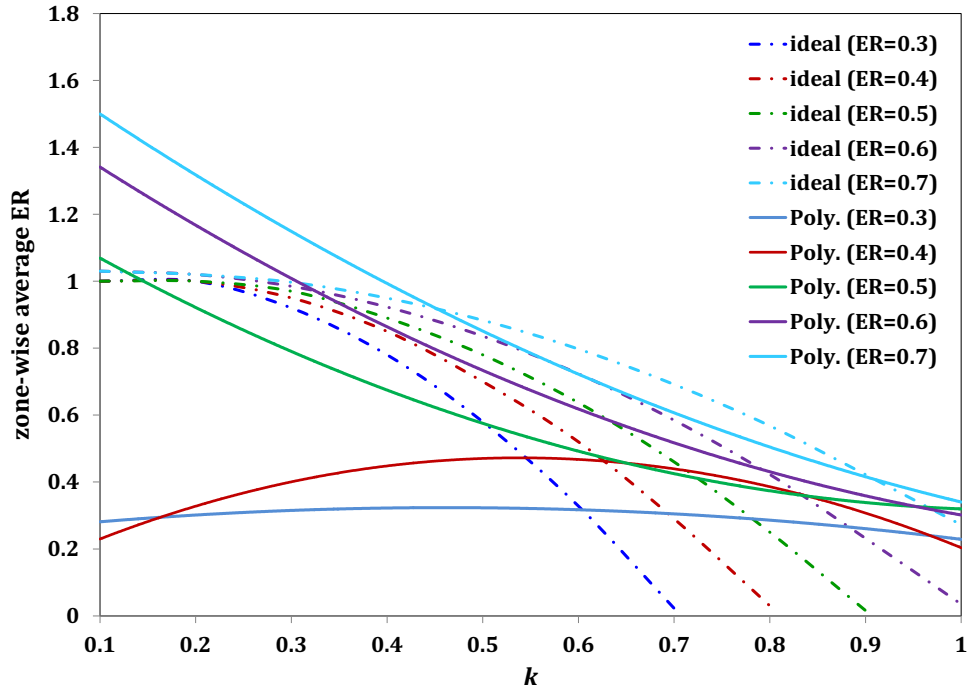


Fig. 21 Comparison of parabolic fits of the baseline case at different ERs

Table 13 Comparison of the parabolic fit equations of the baseline and ideal cases at different ERs

ER	2nd order polynomial fit	Ideal parabolic equation	SI_p	a_a	R-square
0.3	$(x-0.4607)^2 = -3.07(y-0.3233)$	$(x-0)^2 = -0.364(y-1)$	0.3233	0.46	0.36
0.4	$(x-0.53)^2 = -0.7927(y-0.47)$	$(x-0)^2 = -0.5(y-1)$	0.47	0.523	0.65
0.5	$(x-1.06855)^2 = 1.2453(y-0.315)$	$(x-0)^2 = -0.646(y-1)$	0.315	1.06	0.96
0.6	$(x-1.344)^2 = 1.375(y-0.2155)$	$(x-0)^2 = -0.824(y-1)$	0.2155	1.344	0.98
0.7	$(x-1.516)^2 = 1.5(y-0.1631)$	$(x-0)^2 = -1.136(y-1)$	0.1631	1.516	0.97

characteristics, the highest NO_x and the least HC emissions.

- The SI_l of the baseline case, at the overall ER of 0.3, 0.4, 0.5, 0.6 and 0.7 are about 0.022, 0.052, 0.98, 1.5 and 2.1 respectively, which implies that the baseline case has the closest resemblance with the ideal mixture distribution based on the linear variation, at the overall ER of 0.5. Thus, the IMEP of the baseline case, at the overall ER of 0.5, is comparable to its linear mixture distribution counterpart.
- The SI_g of the baseline case, at the overall ER of 0.3, 0.4, 0.5, 0.6 and 0.7, are about 0.32, 0.42, 0.95, 1.25 and 1.39 respectively. Therefore, the baseline case has the closest resemblance with the ideal Gaussian mixture distribution, at the overall ER of 0.5. However, it is not as close as to the ideal Gaussian distribution case as it is to the ideal linear distribution case.
- The SI_p cannot be used for mixture characterization because of its limitation in identifying the richest mixture.

Because of the least number of limitations, the mixture parametrization based on the Gaussian variation is better compared that of the linear and parabolic variations in identifying the location of richest mixture zones and the overall ER in those zones.

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Compliance with Ethical Standards

Conflict of interest On behalf of all authors, the corresponding author states that there is no conflict of interest.

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